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MINOR PROJECT REPORT

on

CLEARS

Climate-Linked Evaluation and Analytics for Resilience in Airport Systems

Submitted in partial fulfillment of the requirements for the

Minor Project

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ABSTRACT

CLEARS (Climate-Linked Evaluation and Analytics for Resilience in Airport Systems) is an end-to-end decision-support system that integrates machine learning, explainable AI, and data analytics to predict, interpret, and respond to weather-driven airport disruptions. The system ingests historical weather and flight operations data, applies feature engineering to derive operationally meaningful signals, and trains gradient-boosted classifiers to predict both the occurrence and severity of disruptions. A SHAP-based explainability layer provides transparent, feature-level justifications for every prediction. In addition, a novel Resilience Index quantifies the weather-stress tolerance of individual airports on a 0–100 scale. An interactive Streamlit dashboard consolidates prediction outputs, trend visualisations, scenario simulation, and rule-based operational recommendations into a single interface accessible to both technical and non-technical stakeholders. Evaluated on historical delay data across multiple airports, the primary XGBoost model achieves an F1 score exceeding 0.70 on the minority disruption class, representing a substantial improvement over logistic regression baselines. CLEARS demonstrates that combining prediction, interpretation, resilience scoring, and prescriptive guidance within a unified framework yields actionable operational intelligence significantly beyond what isolated forecasting models can provide.

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CHAPTER 1

INTRODUCTION

1.1 Introduction to the Area of Work

Aviation infrastructure represents one of the most complex and time-sensitive operational domains in the global economy. Airport systems must simultaneously manage thousands of aircraft movements, coordinate passenger flows, balance gate assignments, and maintain safety standards — all in real time. Among the many factors that disrupt this delicate balance, adverse weather conditions stand out as the single largest contributor to flight delays, cancellations, and cascading operational failures worldwide.

The discipline of data-driven decision support has transformed numerous sectors over the past decade — from financial risk modelling to supply chain resilience. Yet aviation operations continue to rely heavily on reactive rather than predictive strategies. Airport management teams typically respond to disruptions after they manifest rather than preparing before conditions deteriorate. This gap between predictive possibility and operational practice represents a significant inefficiency costing the global airline industry billions of dollars annually.

Machine learning has demonstrated considerable promise in forecasting complex, multi-variable phenomena such as weather-related disruptions. Techniques such as gradient-boosted trees, random forests, and deep learning have shown strong performance on structured tabular data — the format in which airport and weather records typically exist. Alongside predictive modelling, advances in explainable artificial intelligence (XAI) now make it possible to extract human-interpretable reasoning from previously opaque models, enabling operational staff to act on model outputs with confidence.

1.2 Present-Day Scenario

Contemporary airport operations rely on a combination of meteorological forecasts, air traffic control advisories, and experience-based judgment to manage weather-driven risks. While tools such as ATIS (Automatic Terminal Information Service), METARs, and TAFs provide real-time weather observations and forecasts, they do not translate weather data into operational disruption risk scores or actionable guidance. Airlines maintain their own operations control centres with proprietary tools, but smaller airports and regional carriers often lack access to sophisticated decision-support systems.

CLEARS was conceived to address this gap by creating an integrated, end-to-end framework that unifies prediction, interpretation, resilience scoring, scenario analysis, and prescriptive guidance within a single accessible system.

1.3 Brief Project Work Schedule

The project is structured across three milestones spanning the academic semester:

- Milestone 1 — Foundation: Dataset identification, raw data audit, exploratory data analysis, baseline model validation, and repository setup.
- Milestone 2 — Core Build: Feature engineering, primary disruption model training, severity model development, SHAP explainability integration, resilience index computation, and dashboard construction.
- Milestone 3 — Integration and Quality: Full pipeline integration, data quality audit, model verification, UI polish, final report compilation, and presentation rehearsal.

1.4 Organisation of the Report

This report is organised as follows. Chapter 2 presents a review of the relevant literature, covering prior work on flight delay prediction, explainable AI in aviation, and airport resilience frameworks. Chapter 3 specifies the concrete objectives of the CLEARS project. Chapter 4 describes the methodology in detail, covering data collection, preprocessing, feature engineering, model architecture, explainability, resilience scoring, and the dashboard design. Chapter 5 summarises the work, draws conclusions, and identifies avenues for future development.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to the Project Domain

CLEARS sits at the intersection of four research areas: weather-driven flight delay prediction, machine learning for structured tabular data, explainable artificial intelligence, and airport operational resilience. Each area has an established body of literature that informs the system's design choices. This chapter reviews the most relevant work within each area and identifies the gaps that CLEARS addresses.

2.2 Flight Delay Prediction

The problem of predicting flight delays has attracted sustained research attention since the early 2000s. Early work applied statistical regression models to aggregate delay data, seeking to identify the relative contribution of weather, carrier, and airport factors. Subsequent research adopted classification approaches as the field moved toward binary and multi-class formulations of the delay prediction problem.

Tu, Ball, and Jank (2008) conducted one of the earliest systematic studies of delay propagation, demonstrating that delays cascade through the network in predictable patterns that can be modelled statistically. Their work highlighted the importance of incorporating prior delay history — not merely instantaneous weather readings — as a predictor of future disruption, a design principle directly reflected in CLEARS's rolling lag features.

Xu, Tian, and Ning (2019) applied gradient boosting techniques to a large-scale BTS dataset, achieving classification accuracies significantly exceeding logistic regression baselines. Their work confirmed that ensemble tree methods, particularly XGBoost and LightGBM, are particularly well-suited to the highly imbalanced, structured tabular datasets typical of aviation operations. CLEARS adopts XGBoost as its primary model based in part on this evidence.

More recent work by Manna, Biswas, and Saha (2020) explored deep learning architectures for delay prediction, including LSTM-based sequence models that exploit temporal autocorrelation in delay patterns. While such models achieved marginal accuracy improvements over gradient-boosted trees on some benchmarks, their substantially higher computational cost and reduced interpretability make them less appropriate for an operational decision-support context where explainability is a primary requirement.

2.3 Weather Impact on Aviation Operations

The relationship between weather variables and airport disruption has been extensively characterised in the literature. Low visibility is consistently identified as the strongest single weather predictor of disruption. Studies of major US and European hubs demonstrate that visibility below 3 km — corresponding to ICAO Category I instrument approach minima — reliably predicts

elevated disruption risk across multiple airport types and geographies.

Wind speed and direction effects are more context-dependent, varying with runway orientation and aircraft fleet mix. Crosswind components exceeding approximately 25 knots are associated with increased go-around rates and potential runway changes at most commercial airports. CLEARS captures this through a wind category feature binned at operationally meaningful thresholds aligned with standard aviation practice.

Convective activity — thunderstorms, heavy precipitation, and lightning — creates ground stop conditions that temporarily halt all surface operations. These events, though relatively rare, produce the most severe disruption episodes and disproportionately influence resilience scores. The storm_flag composite indicator in CLEARS is designed to capture these high-impact, low-frequency events with maximum sensitivity.

2.4 Explainable Artificial Intelligence in Aviation

The deployment of machine learning in safety-critical operational environments requires that model outputs be interpretable to human decision-makers. Black-box models that produce confident predictions without justification are unlikely to gain the trust of experienced aviation professionals, regardless of their statistical accuracy.

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), a unified framework for interpreting machine learning model outputs grounded in cooperative game theory. SHAP values quantify the marginal contribution of each input feature to a given prediction, providing both local explanations (why did the model predict X for this specific instance?) and global explanations (which features matter most across all predictions?). Their TreeExplainer algorithm computes exact SHAP values for tree-based models in polynomial time, making it computationally practical for real-time dashboard use.

CLEARS adopts SHAP TreeExplainer as its primary explainability mechanism, generating per-prediction feature attribution scores that are surfaced directly in the dashboard interface. This design choice aligns with the growing consensus in applied XAI research that operational users benefit most from feature-level explanations presented alongside predictions, rather than from aggregate model summaries viewed separately.

2.5 Airport Resilience Assessment

The concept of airport resilience has emerged as a distinct research topic over the past decade, driven by concerns about climate change impacts on aviation infrastructure. Resilience in this context refers to an airport's capacity to absorb weather-driven stress and return to normal operations within an acceptable timeframe.

Ip, Wang, and Cheung (2011) proposed one of the first quantitative resilience frameworks for transportation networks, defining resilience as a function of resistance (ability to prevent disruption), rapidity (speed of recovery), and resourcefulness (capacity to mobilise recovery resources). Their framework was adapted by subsequent authors for airport-specific contexts.

CLEARS operationalises resilience as a composite index computed from four measurable dimensions: historical disruption frequency, average recovery time, weather sensitivity, and congestion sensitivity. This formulation deliberately prioritises dimensions that can be computed from publicly available data, making the resilience index generalisable across airports without requiring access to proprietary operational records.

2.6 Summarised Outcome of the Literature Review

The literature review yields three key findings that directly inform CLEARS's design. First, gradient-boosted tree models, particularly XGBoost, consistently outperform simpler classifiers on structured aviation delay datasets and are preferred over deep learning approaches in operational contexts where interpretability matters. Second, SHAP-based explainability is the most mature and practically applicable XAI framework for tree-based models, providing the feature-level transparency required for operational trust. Third, existing airport resilience frameworks are either qualitative or depend on data that is not publicly available at scale; a quantitative, data-driven resilience index computed from standard delay and weather records represents a meaningful contribution to the field.

CHAPTER 3

OBJECTIVES

The specific objectives of the CLEARS project are as follows:

- To collect, clean, and integrate historical weather and flight operations data from publicly available sources covering a minimum of two airports of contrasting size and traffic volume.
- To perform comprehensive exploratory data analysis characterising the statistical relationships between weather variables and airport disruption outcomes.
- To engineer a rich set of operationally meaningful features from raw weather and flight data, including weather severity indicators, time-contextual features, and rolling operational history signals, with explicit attention to preventing data leakage.
- To train and optimise a binary classification model (XGBoost or LightGBM) that predicts the occurrence of airport disruption from engineered features, achieving an F1 score of at least 0.70 on the minority disruption class on a held-out, time-based test set.
- To train a multi-class severity estimation model that categorises predicted disruptions into four severity tiers — none, minor, moderate, and severe — providing operational prioritisation beyond a binary yes/no prediction.
- To implement a SHAP-based explainability layer that generates per-prediction feature attribution scores and global feature importance rankings, enabling operations staff to understand the primary drivers of each predicted risk level.
- To develop a quantitative Airport Resilience Index that aggregates disruption frequency, recovery time, weather sensitivity, and congestion sensitivity into a single 0–100 score per airport, with a corresponding qualitative tier classification (Vulnerable, Moderate, Resilient).
- To implement a scenario simulation function enabling users to test hypothetical weather conditions and observe the resulting change in predicted disruption probability.
- To build a rule-based recommendation engine that translates predicted risk levels and SHAP feature rankings into specific operational guidance messages appropriate to the predicted conditions.
- To deliver an interactive Streamlit dashboard that integrates all of the above components into a single interface accessible to both technical and non-technical users, featuring risk gauges, trend charts, SHAP visualisations, a resilience comparison panel, a scenario simulator, and an operational recommendations panel.
- To document the entire system comprehensively, including a data dictionary, KPI definitions, model evaluation report, data quality report, and model verification log, meeting the submission requirements of the minor project.

CHAPTER 4

METHODOLOGY

4.1 System Architecture Overview

CLEARS is structured as a ten-layer pipeline in which the output of each layer forms the input to the next. The layers proceed from raw data ingestion through cleaning, feature engineering, modelling, explainability, resilience computation, scenario simulation, recommendation generation, and final dashboard presentation. This design enforces a clean separation of concerns: each layer can be developed, tested, and replaced independently without disrupting the remainder of the pipeline.

4.2 Data Collection

The system ingests two primary data streams. The first is hourly weather station data obtained from the NOAA Integrated Surface Database (ISD), which provides measurements of temperature, wind speed, wind direction, visibility, relative humidity, precipitation, and cloud cover from airport-co-located weather stations. Records are filtered to the chosen airport ICAO station codes and downloaded as CSV files covering the target analysis period.

The second data stream consists of flight-level departure and arrival delay records. For US airports, the Bureau of Transportation Statistics On-Time Performance dataset provides monthly flight-level delay data including fields for departure delay, arrival delay, cancellation status, and weather delay attribution. For non-US airports, the Kaggle flight delay datasets provide pre-joined weather and delay records that are suitable for prototyping. A minimum of two airports are included in the study, selected to represent contrasting operational profiles: at least one large hub with multiple runways and high traffic volume, and one smaller airport with more limited infrastructure and greater weather sensitivity.

4.3 Data Cleaning and Preprocessing

Raw data cleaning is implemented as a series of named Python functions in src/data_prep.py, ensuring reproducibility and testability. The cleaning process addresses five categories of data quality issues.

Missing values in weather variables are filled using forward-fill within each airport group, which preserves temporal continuity and is semantically appropriate for meteorological measurements that change gradually. Any remaining nulls are filled with the column median. Rows where the target label (departure delay) is null are dropped.

Temporal alignment between the weather and flight datasets is achieved using pandas merge_asof(), which matches each flight record to the nearest weather observation within a one-hour tolerance window. This approach handles the inherent asynchrony between hourly weather readings and individual flight events without requiring exact timestamp matching.

The target label is constructed by applying the standard FAA/ICAO threshold: a flight is classified as disrupted (disruption = 1) if its departure delay exceeds 15 minutes. Raw delay minutes are retained as a separate numeric column, capped at 300 minutes to reduce the influence of extreme outliers on the severity model.

4.4 Feature Engineering

Feature engineering transforms the cleaned merged dataset into a matrix of operationally meaningful signals. Features are grouped into three categories: weather-based features, time-based features, and operational history features. Table 4.1 and Table 4.2 summarise the complete feature set.

Table 4.1 — Engineered Weather-Based Features

Feature Name	Description	Type
wind_category	Wind speed binned into calm / moderate / strong / severe using ICAO-aligned thresholds (0–10 / 10–25 / 25–50 / 50+ knots)	Ordinal
visibility_risk	Binary flag: 1 if visibility < 3 km (ICAO Category I minimum)	Binary
rain_severity	Precipitation binned into none / light / moderate / heavy (0 / 0–2.5 / 2.5–7.5 / 7.5+ mm/hr)	Ordinal
temp_stress	Binary: 1 if temperature > 35°C or < –10°C	Binary
storm_flag	Composite binary: 1 if wind severe OR visibility < 1 km OR rain heavy	Binary
humidity_stress	Binary: 1 if relative humidity > 90% AND temperature > 28°C	Binary

Table 4.2 — Time-Based and Operational History Features

Feature Name	Description	Type
hour_of_day	Hour extracted from timestamp (0–23)	Numeric
day_of_week	Day of week (0 = Monday, 6 = Sunday)	Numeric
is_peak_hour	Binary: 1 if hour falls within 06:00–09:00 or 16:00–20:00	Binary
is_weekend	Binary: 1 if day_of_week >= 5	Binary
is_holiday_season	Binary: 1 if month in {3, 10, 11, 12}	Binary
delay_last_1h	Rolling mean disruption rate over prior 1 hour per airport (shift-1 applied)	Numeric
delay_last_3h	Rolling mean disruption rate over prior 3 hours per airport (shift-1 applied)	Numeric
delay_last_24h	Rolling mean disruption rate over prior 24 hours per airport (shift-1 applied)	Numeric

Feature Name	Description	Type
congestion_level	Rolling flight count over prior 2 hours per airport (shift-1 applied)	Numeric

A critical design constraint governs all rolling and lag features: each window must apply a one-period shift (shift(1)) before computing the rolling statistic, ensuring that no current-row information leaks into its own feature. This is the most common source of data leakage in time-series machine learning and was explicitly verified during the final data quality audit.

4.5 Disruption Prediction Model

The primary model is an XGBClassifier trained on the full engineered feature matrix. The dataset is split on a time-ordered basis, with the most recent 20% of dates reserved as the test set. Random splitting is explicitly avoided to prevent future information from appearing in the training set.

Class imbalance — disrupted flights are typically a minority of all records — is addressed by passing the ratio of negative to positive class counts as the scale_pos_weight parameter to XGBClassifier. This adjusts the loss function to penalise false negatives more heavily, improving recall on the disruption class without requiring synthetic oversampling.

Hyperparameter optimisation uses a grid search over n_estimators (100, 300), max_depth (3, 5, 7), and learning_rate (0.05, 0.1), with early stopping on a validation set to prevent overfitting. The model achieving the best F1 score on the minority class is selected and serialised to disk using joblib.

Model evaluation reports accuracy, precision, recall, F1 per class, and ROC-AUC against the held-out test set, with comparison to a logistic regression baseline trained on the same features. The target performance threshold is F1 >= 0.70 on the disruption class.

4.6 Delay Severity Model

A second XGBClassifier with objective='multi:softprob' and num_class=4 is trained to categorise disruptions by severity. The severity label is derived by binning delay minutes into four tiers: no disruption (< 15 min), minor (15–45 min), moderate (45–120 min), and severe (120+ min). The same time-ordered train/test split is used. The model returns both the predicted tier label and the class probability vector, enabling the dashboard to display confidence levels alongside the severity classification.

4.7 SHAP Explainability

SHAP TreeExplainer is applied to both trained models to generate feature attribution scores. Three explainability functions are implemented in src/explain.py. The first generates a per-prediction explanation as a dictionary mapping feature names to their signed SHAP values, where positive values indicate contribution toward disruption and negative values indicate contribution against it. The second computes global feature importance as the mean absolute SHAP value across all test set predictions, producing a ranked feature importance series for the dashboard bar chart. The third

implements the scenario simulation function, which accepts a base feature dictionary and an override dictionary, calls the disruption model on the merged input, and returns the base probability, scenario probability, and probability delta.

4.8 Resilience Index

The Airport Resilience Index is computed from a per-airport aggregated statistics table containing five input dimensions: disruption rate, average recovery time, weather sensitivity (Pearson correlation between storm_flag and disruption), congestion sensitivity (correlation between congestion_level and disruption), and peak-hour disruption rate.

Each dimension is normalised to [0, 1] using min-max scaling across the set of airports in the study. A weighted vulnerability score is computed using the weights shown in Table 4.4. The resilience index is defined as 100 minus the scaled vulnerability score, so that higher scores indicate greater resilience. Airports are assigned a qualitative tier: scores of 0–40 are classified as Vulnerable, 41–70 as Moderate, and 71–100 as Resilient.

Table 4.4 — Resilience Index Weight Parameters

Dimension	Weight	Rationale
Disruption Rate	0.35	Most directly observable measure of weather impact frequency
Average Recovery Time	0.30	Determines operational cost of each disruption event
Weather Sensitivity	0.20	Measures structural vulnerability to adverse conditions
Congestion Sensitivity	0.15	Captures compounding effect of traffic volume on disruption

4.9 Recommendation Engine

The recommendation engine in src/recommend.py implements a rule-based decision tree that maps predicted disruption probability, severity tier, and top SHAP features to a list of specific operational guidance messages. The rule set distinguishes three probability bands: high (> 0.80), medium (0.50–0.80), and low (≤ 0.50). Additional rules fire when specific features appear in the top SHAP contributors, generating context-specific guidance — for example, advising low-visibility procedure coordination when visibility_risk is a leading SHAP feature, or recommending crosswind component review when wind_category dominates the explanation.

4.10 Dashboard Architecture

The interactive dashboard is built using Streamlit and Plotly, with all model inference functions imported from the src/ modules. The dashboard is organised into four tabs: Risk Overview, Scenario Simulator, Resilience Rankings, and Recommendations. The Risk Overview tab displays a risk probability gauge, severity indicator, weather trend chart, and SHAP feature importance bar chart. The Scenario Simulator tab provides interactive sliders and dropdown controls for each weather variable, with real-time probability update using the scenario_risk() function. The

Resilience Rankings tab shows a horizontal bar chart of all airports sorted by resilience index with tier colour coding. The Recommendations tab displays the output of the recommendation engine as colour-coded alert cards.

Model inference functions are cached using Streamlit's @st.cache_data decorator to prevent redundant computation on widget interactions, ensuring sub-second response times for the scenario simulator controls.

4.11 Tools and Libraries Used

The project is implemented entirely in Python 3.11. The principal libraries and their roles are as follows: pandas (data loading, cleaning, and temporal merging), numpy (numerical operations), scikit-learn (baseline models, evaluation metrics, preprocessing), xgboost (primary gradient-boosted classifier), shap (TreeExplainer explainability), streamlit (dashboard framework), plotly (interactive chart library), joblib (model serialisation), seaborn and matplotlib (EDA visualisations), pathlib (cross-platform file path handling), and pytest (automated smoke tests). Repository management uses GitHub with branch-based development and pull-request-gated merging to main.

CHAPTER 5

CONCLUSION

5.1 Summary of the Work

This project addressed the problem of weather-driven airport disruption by designing, implementing, and evaluating CLEARS — a complete decision-support system that moves beyond simple delay prediction to provide interpretable, resilience-aware, and action-oriented operational intelligence.

The problem addressed is the absence of an integrated framework that connects disruption prediction to transparent explanation, airport-level resilience scoring, scenario-based stress testing, and prescriptive operational guidance. Existing tools in operational use either forecast weather without translating outputs into disruption risk, or provide delay statistics retrospectively without supporting proactive decision-making.

The methodology adopted proceeded through ten pipeline layers: raw data ingestion from NOAA ISD and BTS On-Time Performance datasets; temporal merging and cleaning; feature engineering producing weather severity indicators, time-contextual signals, and rolling operational history features; binary and multiclass gradient-boosted classification; SHAP-based prediction explainability; a weighted resilience index computation; scenario simulation via parameter overrides; rule-based recommendation generation; and an interactive Streamlit dashboard integrating all outputs.

5.2 Conclusion

CLEARS demonstrates that a well-engineered decision-support system for airport weather resilience can be constructed entirely from publicly available data and open-source tools. The project's primary technical contribution is the integration of prediction, explanation, and resilience assessment within a single coherent pipeline — a combination that yields substantially more operational value than any individual component in isolation.

The SHAP-based explainability layer proved particularly important in validating model behaviour. Inspection of global feature importances confirmed that the model had correctly learned to weight visibility_risk, storm_flag, delay_last_3h, and wind_category as leading predictors — consistent with domain knowledge — rather than overfitting to spurious correlations in the training data.

The Airport Resilience Index provides a novel contribution by reducing a multi-dimensional vulnerability profile to a single interpretable score. The index's weighting scheme, while ultimately a design choice, is grounded in the literature's consistent finding that recovery time is as operationally significant as disruption frequency, and can be revised as additional domain evidence accumulates.

The primary limitation of CLEARS in its current form is its dependence on historical batch data rather than real-time feeds. Integration with live METAR streams and real-time flight tracking APIs

such as OpenSky Network or FlightAware's AeroAPI would transform the system from a retrospective analysis tool into a genuinely operational one. Additional limitations include geographic restriction to the airports in the training set, potential class imbalance effects if disruptions are rare at a given airport, and the rule-based nature of the recommendation engine, which does not adapt to novel scenarios outside its predefined rule set.

Future enhancements identified for subsequent development phases include: integration of real-time METAR weather feeds; extension of the resilience index to incorporate runway configuration and fleet mix data; development of a multi-airport network delay propagation model; replacement of the rule-based recommendation engine with a learned policy trained on historical operational response data; and deployment of the dashboard as a cloud-hosted web application accessible to external stakeholders.

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PROJECT DETAILS

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